

Package ‘CensRCalc’

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Title Analytical Censoring Parameter Determination

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BugReports <https://github.com/iden-project-uas-darmstadt/CensRCalc/issues>

Description Provides calculations for the expected censoring proportion under independent censoring to determine a censoring distribution parameter that achieves a target censoring proportion in simulation studies. The expected censoring proportion is evaluated by numerical integration over time and covariates, with support for administrative censoring and accrual periods.

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CensRCalc-package	<i>CensRCalc: Analytical Censoring Parameter Determination</i>
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Description

Provides calculations for the expected censoring proportion under independent censoring to determine a censoring distribution parameter that achieves a target censoring proportion in simulation studies. The expected censoring proportion is evaluated by numerical integration over time and covariates, with support for administrative censoring and accrual periods.

Details

Let the event time be $T^* \mid \mathbf{X} \sim F_{T^* \mid \mathbf{X}}$ with survival function $S_{T^* \mid \mathbf{X}} = 1 - F_{T^* \mid \mathbf{X}}$ and the random censoring time be $C_{\text{rnd}} \mid \lambda_{C_{\text{rnd}}} \sim F_{C_{\text{rnd}} \mid \lambda_{C_{\text{rnd}}}}$ with survival function $S_{C_{\text{rnd}} \mid \lambda_{C_{\text{rnd}}}} = 1 - F_{C_{\text{rnd}} \mid \lambda_{C_{\text{rnd}}}}$, where $\lambda_{C_{\text{rnd}}}$ is the random censoring parameter of interest. Administrative censoring occurs at time τ_{adm} . Accrual takes place over $[0, \tau_{\text{acc}}]$ where the random accrual time is $R \sim U(0, \tau_{\text{acc}})$. The equivalent random administrative censoring time after accrual is $C_{\text{adm}} = \tau_{\text{adm}} - R$. Hence, $C_{\text{adm}} \sim U(\tau_{\text{adm}} - \tau_{\text{acc}}, \tau_{\text{adm}})$.

Under these assumptions, the probability of censoring dependent on $\lambda_{C_{\text{rnd}}}$ and the covariate vector $\mathbf{X}$ is

$$\begin{aligned}
 P(\delta = 0 \mid \mathbf{X}, \lambda_{C_{\text{rnd}}}) = & \int_0^{\tau_{\text{adm}} - \tau_{\text{acc}}} S_{T^* \mid \mathbf{X}}(c_{\text{rnd}}) f_{C_{\text{rnd}} \mid \lambda_{C_{\text{rnd}}}}(c_{\text{rnd}}) dc_{\text{rnd}} + \\
 & \int_{\tau_{\text{adm}} - \tau_{\text{acc}}}^{\tau_{\text{adm}}} S_{T^* \mid \mathbf{X}}(c_{\text{rnd}}) S_{C_{\text{adm}}}(c_{\text{rnd}}) f_{C_{\text{rnd}} \mid \lambda_{C_{\text{rnd}}}}(c_{\text{rnd}}) dc_{\text{rnd}} + \\
 & \int_{\tau_{\text{adm}} - \tau_{\text{acc}}}^{\tau_{\text{adm}}} S_{T^* \mid \mathbf{X}}(C_{\text{adm}}) S_{C_{\text{rnd}} \mid \lambda_{C_{\text{rnd}}}}(C_{\text{adm}}) f_{C_{\text{adm}}}(C_{\text{adm}}) dC_{\text{adm}}
 \end{aligned}$$

Next we take the expectation over the covariate distribution to get the expected censoring proportion:

$$P(\delta = 0 \mid \lambda_{C_{\text{rnd}}}) = E_X \left[P(\delta = 0 \mid \mathbf{X}, \lambda_{C_{\text{rnd}}}) \right].$$

This is implemented in `estimate_cens_prop()`. To obtain the censoring parameter $\lambda_{C_{\text{rnd}}}$ that achieves a target censoring proportion p_C , we solve the root finding problem

$$P(\delta = 0 \mid \lambda_{C_{\text{rnd}}}) - p_C = 0$$

Special cases include no accrual time ($\tau_{\text{acc}} = 0$) and no administrative censoring ($\tau_{\text{adm}} = \infty$). See the package vignette `vignette("full_details", package = "CensRCalc")` for details.

Numerically, the integrals are evaluated using adaptive cubature (`calculus::integral()`), which was extended to handle discrete and mixed discrete/continuous covariates.

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See Also

Useful links:

- <https://iden-project-uas-darmstadt.github.io/CensRCalc>
- Report bugs at <https://github.com/iden-project-uas-darmstadt/CensRCalc/issues>

cens_prop_fun-methods *Print and plot methods for censoring proportion functions*

Description

Print and plot methods for censoring proportion functions

Usage

```
## S3 method for class 'cens_prop_fun'
print(x, ...)

## S3 method for class 'cens_prop_fun'
plot(x, n = 100, tau = NULL, ...)
```

Arguments

x	A "cens_prop_fun" object.
...	Additional arguments.
n	Number of evaluation points.
tau	Optional time point for evaluating the expected censoring proportion. If NULL, the censoring proportion is evaluated overall, otherwise it is evaluated at the specified time point.

Value

The input object, invisibly.

estimate_cens_prop	<i>Calculate expected censoring proportion</i>
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Description

Evaluates the expected censoring proportion under independent censoring by integrating survival and censoring functions over time and covariates. Administrative censoring and accrual periods are supported.

Usage

```
estimate_cens_prop(
  event_survival,
  covariate_density = NULL,
  covariate_bounds = NULL,
  cens_survival = function(t, lambda_c) {
    1 - stats::pexp(t, rate = lambda_c)
  },
  cens_density = function(t, lambda_c) {
    stats::dexp(t, rate = lambda_c)
  },
  time_admin_cens = t_max,
  time_accrual = t_min,
  target = "lambda_c",
  target_bounds = c(1e-05, 5),
  t_min = 0,
  t_max = Inf,
  abs_tol = 1e-06
)
```

Arguments

event_survival	Function. Event-time survival function $S_{T^* \mathbf{X}}(t)$. The first argument must be t and further arguments are considered covariates. Must be vectorized in t .
covariate_density	Optional function. Joint density of the covariate vector over covariate_bounds. If provided, it must integrate to one over the specified domain. If NULL, no covariates are integrated.
covariate_bounds	Named list or NULL. Bounds for covariates used both to integrate covariate_density. Use numeric length-2 vectors for continuous bounds and lists of numeric values for discrete variables. Names must match the non- t arguments of event_survival. Must be NULL when covariate_density is NULL.
cens_survival	Function. Censoring-time survival function $S_{C_{\text{rnd}} \lambda_{C_{\text{rnd}}}}(t)$ with first argument t and the second argument named as in target. Must be vectorized in t .
cens_density	Function. Censoring-time density $f_{C_{\text{rnd}} \lambda_{C_{\text{rnd}}}}(t)$ with first argument t and the second argument named as in target. Must be vectorized in t and be consistent with cens_survival.
time_admin_cens	Numeric scalar. Administrative censoring time τ_{adm} . Use Inf for no administrative censoring.
time_accrual	Numeric scalar. Accrual time span τ_{acc} . Use 0 for no accrual.
target	Character scalar. Name of the censoring parameter inside cens_survival and cens_density (e.g., "lambda_c").
target_bounds	Numeric length-2 vector. Lower and upper bounds of the validation interval for the parameter named in target.
t_min	Numeric scalar. Left boundary of the time support used for validation and numerical checks. Must be non-negative.
t_max	Numeric scalar. Right boundary of the time support used for validation and numerical checks. Use Inf for an unbounded right tail.
abs_tol	Numeric scalar. Absolute tolerance used for integration (on the probability scale). Must be positive and not too large (e.g., < 0.1).

Details

This function returns the expected censoring proportion over the covariates $P(\delta = 0 \mid \lambda_{C_{\text{rnd}}}) = E_X \left[P(\delta = 0 \mid \mathbf{X}, \lambda_{C_{\text{rnd}}}) \right]$ obtained by numerical integration of survival and censoring functions over time and finally integrating over the covariate distributions. The mathematical background and the full expression for $P(\delta = 0 \mid \lambda_{C_{\text{rnd}}})$ are provided in the package description, see [CensRCalc](#).

Functions supplied for time must be vectorized in their first argument (time). The survival functions and densities of the event time and censoring time variables, respectively, must be consistent, i.e., they correspond to the same distribution.

Value

A callable function of two numeric scalars, one named by `target` and the other `tau`, which returns the expected censoring proportion under the specified model components. When `tau` is specified, the expected censoring proportion is evaluated at the time point `tau` instead of overall time.

The returned function has class "cens_prop_fun" and supports S3 methods such as `print()` and `plot()` for summarizing and visualizing the censoring model.

Examples

```
# Exponential event time without covariates
event_surv <- function(t) exp(-0.2 * t)

cens_surv <- function(t, lambda_c) exp(-lambda_c * t)
cens_dens <- function(t, lambda_c) lambda_c * exp(-lambda_c * t)

f_obj <- estimate_cens_prop(
  event_survival = event_surv,
  covariate_density = NULL,
  covariate_bounds = NULL,
  cens_survival = cens_surv,
  cens_density = cens_dens,
  time_admin_cens = 10,
  time_accrual = 0,
)

print(f_obj)
f_obj(0.3)
f_obj(0.3, tau = 5)
```

find_cens_param

Determine a censoring parameter for a target censoring proportion

Description

Solves for the parameter of a random censoring distribution such that the expected censoring proportion equals a predefined target proportion. This is designed for simulation studies where target censoring proportions are required. Administrative censoring and accrual periods are supported.

Usage

```
find_cens_param(
  cens_prop,
  event_survival,
  covariate_density = NULL,
  covariate_bounds = NULL,
  cens_survival = function(t, lambda_c) {
    1 - stats::pexp(t, rate = lambda_c)
  },
```

```

    cens_density = function(t, lambda_c) {
      stats::dexp(t, rate = lambda_c)
    },
    tau = NULL,
    time_admin_cens = t_max,
    time_accrual = t_min,
    target = "lambda_c",
    target_bounds = c(1e-05, 5),
    t_min = 0,
    t_max = Inf,
    abs_tol = 1e-06
  )

```

Arguments

cens_prop	Numeric scalar in $[0, 1]$. Expected target censoring proportion in the simulated data.
event_survival	Function. Event-time survival function $S_{T^* \mathbf{X}}(t)$. The first argument must be t and further arguments are considered covariates. Must be vectorized in t .
covariate_density	Optional function. Joint density of the covariate vector over <code>covariate_bounds</code> . If provided, it must integrate to one over the specified domain. If NULL, no covariates are integrated.
covariate_bounds	Named list or NULL. Bounds for covariates used both to integrate <code>covariate_density</code> . Use numeric length-2 vectors for continuous bounds and lists of numeric values for discrete variables. Names must match the non- t arguments of <code>event_survival</code> . Must be NULL when <code>covariate_density</code> is NULL.
cens_survival	Function. Censoring-time survival function $S_{C_{\text{rnd}} \lambda_{C_{\text{rnd}}}}(t)$ with first argument t and the second argument named as in <code>target</code> . Must be vectorized in t .
cens_density	Function. Censoring-time density $f_{C_{\text{rnd}} \lambda_{C_{\text{rnd}}}}(t)$ with first argument t and the second argument named as in <code>target</code> . Must be vectorized in t and be consistent with <code>cens_survival</code> .
tau	Numeric scalar. Optional time point at which the expected censoring proportion is evaluated. If NULL, the expected censoring proportion is evaluated overall, otherwise it is evaluated at the specified time point.
time_admin_cens	Numeric scalar. Administrative censoring time τ_{adm} . Use <code>Inf</code> for no administrative censoring.
time_accrual	Numeric scalar. Accrual time span τ_{acc} . Use <code>0</code> for no accrual.
target	Character scalar. Name of the censoring parameter inside <code>cens_survival</code> and <code>cens_density</code> (e.g., "lambda_c").
target_bounds	Numeric length-2 vector. Lower and upper bounds of the search interval for the parameter named in <code>target</code> .
t_min	Numeric scalar. Left boundary of the time support used for validation and numerical checks. Must be non-negative.

t_max	Numeric scalar. Right boundary of the time support used for validation and numerical checks. Use Inf for an unbounded right tail.
abs_tol	Numeric scalar. Absolute tolerance used for integration (on the probability scale). Must be positive and not too large (e.g., < 0.1).

Details

The parameter $\lambda_{C_{\text{rnd}}}$ is determined by solving the root finding problem

$$P(\delta = 0 \mid \lambda_{C_{\text{rnd}}}) - p_C = 0$$

where $P(\delta = 0 \mid \lambda_{C_{\text{rnd}}})$ is the expected censoring proportion over the covariate distributions and $p_C \in [0, 1]$ is the target censoring proportion. The expected censoring proportion $P(\delta = 0 \mid \lambda_{C_{\text{rnd}}}) = E_X \left[P(\delta = 0 \mid \mathbf{X}, \lambda_{C_{\text{rnd}}}) \right]$ is evaluated by [estimate_cens_prop\(\)](#) and its mathematical background is detailed in the package description ([CensRCalc](#)). A bracketing root finder is used with the interval given by `target_bounds`.

Convergence depends on the chosen bounds covering a sign change of the objective function. The tolerance `abs_tol` is used for both integration accuracy and the root-finding convergence criterion.

Progress reporting is supported via the **progressr** package. When a `progressr` handler is enabled (for example, `progressr::with_progress()`), the iterations of the bracketing solver emit progress updates.

Value

A list with components:

- `parameter`: the value of the censoring parameter.
- `cens_prop`: the achieved expected censoring proportion at the calculated parameter.

See Also

[estimate_cens_prop\(\)](#) for the analytical evaluation of the expected censoring proportion, and [CensRCalc](#) for mathematical details.

Examples

```
# Target 30% censoring under exponential models without covariates
event_surv <- function(t) exp(-0.2 * t)

sol <- find_cens_param(
  cens_prop = 0.3,
  event_survival = event_surv
)

sol$parameter
```

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